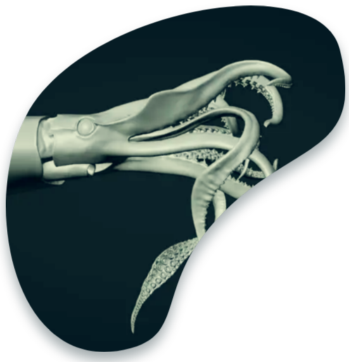

Stock Assessment of Jumbo Flying Squid

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Category: Working paper summary

Table of contents

1 Document

A technical paper by CALAMASUR (Roa-Ureta and Wiff 2025), presented as an observer paper (SC13-Obs03) to the 13th Meeting of the SPRFMO Scientific Committee (8–13 September 2025, Wellington, New Zealand). Dated August 7, 2025.

2 Subject

A region-wide stock assessment of jumbo squid (*Dosidicus gigas*) across the South East Pacific (SEP) — the world's largest invertebrate fishery (over 1 million tonnes in recent years, 15th largest of all fisheries). It is fished in Ecuadorian, Peruvian, and Chilean EEZs plus international high seas (currently all Chinese vessels; Chinese Taipei operated until 2020, South Korea until 2019).

3 Methods

A two-stage, environment-driven approach spanning 1990–2024:

- **Stage 1** fits multi-annual, multi-fleet *generalized depletion models* to a monthly database (Jan 2012–Dec 2024) of total catch, fishing effort, and mean weight, for three fleets: Peruvian, Chilean, and Asian. The database covers >99% of regional landings.
- **Stage 2** uses a *Pella-Tomlinson surplus production model* driven by El Niño/La Niña cycles (NOAA's Oceanic Niño Index, 10 transitions 1990–2024), testing 16 hypotheses about how carrying capacity, production symmetry, and intrinsic growth rate change with the environment.

4 Key results

- Depletion models fit Chilean and Asian data well, Peruvian data less so (fits ~90% of points).
- Natural mortality estimated at 0.17/month (2.0/year), consistent with the species' short life cycle.
- Biomass fluctuates widely (maxima an order of magnitude above minima) but shows **no sustained downward trend**; highest peaks were before 2019.
- Exploitation rates have risen since 2020 but remain **mostly below 40%** (peaked ~40% in 2018, now ~20–30%).
- The best-supported hypothesis is that only the intrinsic growth rate r varies with the environment — higher during El Niño (2.93/yr) than La Niña (0.48/yr).

5 Main conclusion

The stock is **not overfished**, though it **may be undergoing overfishing**. Because the stock has entered a regime of strong environment-driven fluctuations, MSY and B_{MSY} are judged inadequate reference points, and a standard Kobe plot cannot be built. During El Niño years the model even shows negative surplus production. Estimates of sustainable harvest rates (mean latent productivity) remain statistically imprecise.

6 Way forward

Develop models with 4 fleets and 3 natural-mortality zones (Peru, Chile, high seas), and move the surplus-production model to a continuous, monthly ONI-driven time step. Better environmental reference points are needed to properly judge exploitation status.

References

Roa-Ureta, Rubén H., and Rodrigo Wiff. 2025. *Stock Assessment of the Jumbo Squid (Dosidicus gigas) in the Eastern Pacific with Multi-Annual Multi-Fleet Generalized Depletion Models Combined with Environmental Surplus Production Models (1990–2024).* Scientific Committee Document Nos. SC13-Obs03. South Pacific Regional Fisheries Management Organisation (SPRFMO).